

Caner Deric

dericilab.live caner@dericilab.live [in canerderici](https://www.linkedin.com/in/canerderici) [cderici](https://github.com/cderici) [UT, USA](#) [US Citizen](#) [extended version](#)

Technical Skills

Areas of Experience:	Compilers · Distributed Systems · Machine Learning
Languages:	C++ · Go · Python · Racket/Scheme · LLVM · Java · SQL · JavaScript
Cloud:	Kubernetes · AWS · GCE · Terraform · LXD · Docker
Workflows & Productivity:	Linux · Neovim · tmux · VSCode · OpenCode · Codex · Git · GH Actions · Obsidian
APIs, DB & Misc:	REST · gRPC · GraphQL · OpenAPI · FastAPI · DQLite · MongoDB · PostgreSQL · CI/CD

Education

Ph.D., Indiana University, Bloomington , Computer Science, Compilers	2015 – 2025
M.Sc., Boğaziçi University , Computer Science, Machine Learning	2012 – 2014
B.Sc., Bilgi University , Computer Science	2005 – 2010

Experience

Canonical USA	REMOTE, US
SWE II (L4 - IC3) · Distributed Orchestration at Scale	2021 – 2024
- Developed and maintained Juju, an eventually consistent distributed orchestration system used by ~200 companies globally, capable of handling 1000+-node workloads with 99.9% availability on any infrastructure (Kubernetes or otherwise) across various cloud providers (e.g., AWS, GCE). All in Go. - Improved reliability and fault tolerance by implementing machine provision services on relational DQLite state back-end, migrating from NoSQL MongoDB (e.g., sample PR). - Owned client libraries for three years—python-libjuju, Terraform Juju Provider; doubled active users and maintained a steady release cadence. - Took part in roadmap planning, coordinated cross-team work; mentored junior engineers, and created structured bootcamp process to improve hiring and onboarding.	
Indiana University	IN, US
PhD Work · Research Assistant · Course Instructor	2015 – 2021
- Independently took an ambiguous, uncharted compiler problem to a working product; built the first-ever tracing JIT compiler that is a full-scale runtime for a self-hosting, production-grade language. Conducted a full performance investigation and designed new optimization algorithms (see PhD dissertation). - Taught data structures & algorithms, compilers, virtual machines, and domain specific languages.	
Asseco SEE Group	
Software Engineer	2010-2012
International software company developing virtual payment systems for e-commerce platforms. I developed and delivered 3 virtual point-of-sale projects in 2 years. Used Java, Apache Tomcat, Spring, Mercurial, Jira.	

Selected Projects

[Pycket: A tracing JIT compiler for full-scale bootstrapping functional language](#)

Built the first tracing JIT capable of running a self-hosting language on a meta-tracing JIT backend, and worked on it [for over five years](#) during my PhD. Created a new IR and implemented [performance analysis tools](#), run-time optimizations, and [formalisms](#) to improve throughput. Added code-gen for the FFI layer, engines, and meta-continuations for user threads. Used Python, C++, and Racket.

[Around the World of Compilers](#)

Designed and implemented a growing collection of experimental compilers, one for each letter of the alphabet. Built shared frontend pipelines targeting LLVM IR, MLIR, NVPTX, and LLVM MIR. Explored MLIR affine/linalg lowering, CUDA/NVPTX code-gen, MLIR.smt + Z3 runtime verification, and out-of-tree LLVM MachineScheduler plugins for MIR scheduling experiments, as well as LLVM ORC JIT compilation. In C++23, LLVM/MLIR 20.

[Rax: A full-stack nanopass compiler from Racket to x86_64 ISA](#)

Implemented all the passes (closure conversion, register allocation, code-gen, etc.), along with garbage collection. Developed optimizations, such as inlining, loop-invariant code motion, and proper tail calls. In Racket and C++14.

[Terraform Juju Provider](#)

A Terraform provider that enables integration with [Juju](#) while managing Terraform environments. I implemented new resources and features (e.g., [manual provisioning on AWS](#)), migrated the provider from the sdk2 to the provider framework (e.g., [sample PR](#)), and maintained release cadence of new versions. All in Go.

FARS: Functional Automated Reasoning System

A resolution/refutation theorem prover, for expressions in first-order predicate logic with equality. Used binary paramodulation, and forward and backward subsumption for equational deduction. In Racket.

HazirCevap (Witty): A closed-domain question-answering system for high school students

Government-funded large-scale question-answering system. M.Sc. thesis on NLP. Led the R&D team (3 faculty members, 4 grad students). Developed a Hidden Markov Random Field model for question analysis, and relevance metrics for information retrieval and response generation (see publications). In Python and JavaScript.

Selected Publications

- Flatt M., Derici C. Dybvig R. K., Keep A. et. al. "Rebuilding racket on chez scheme (experience report)", ICFP'19
- Derici C. et. al. "A closed-domain question answering framework using reliable resources to assist students" Natural Language Engineering'18
- Derici C. et. al. "Question analysis for a closed domain question answering system", CICLING'15
- Derici C. et. al. "Rule-based focus extraction in Turkish question answering systems", SIU'14
- Başar R. E., Derici C., and Şenol Ç. "World With Web: A compiler from world applications to JavaScript". Technical Report, Scheme and Functional Programming Workshop'09

Awards & Scholarships

- Scholarship and award for a project on teaching natural languages to hearing impaired, 2014.
 - Full Scholarship for PhD, 2015-2025
 - Full Scholarship for MSc, 2012
 - Full Scholarship for BSc, 2005-2010
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